THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS · EASY LEVEL CHAPTER TEST SET 3

Chapter 5: **Circles** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The number of circles passing through two given points is: 1

(a) exactly two **(b)** infinitely many **(c)** exactly one **(d)** none

2. If PQ is a diameter and R lies on the circle, then $\angle PRQ =$ 1

(a) 90° **(b)** 120° **(c)** 60° **(d)** 135°

3. The radius of a circle of diameter 30 cm is (in cm): 1

(a) 16 **(b)** 17 **(c)** 15 **(d)** 14

4. The perpendicular from the centre of a circle to a chord of length 22 cm bisects it. Each half is (in cm): 1

(a) 10 **(b)** 12 **(c)** 13 **(d)** 11

5. The angle subtended by an arc at the centre is 110°. The angle it subtends at a point on the remaining part of the circle is: 1

(a) 55° **(b)** 65° **(c)** 75° **(d)** 45°

6. An arc subtends 62° at a point on the circle. At the centre it subtends: 1

(a) 134° **(b)** 114° **(c)** 144° **(d)** 124°

7. $ABCD$ is a cyclic quadrilateral with $\angle A = 110^\circ$. Then $\angle C =$ 1

(a) 70° **(b)** 80° **(c)** 60° **(d)** 90°

8. Equal chords of a circle subtend ___ angles at the centre. 1

(a) complementary **(b)** unequal **(c)** supplementary **(d)** equal

9. The radius of a circle of diameter 36 cm is (in cm): 1

(a) 18 **(b)** 19 **(c)** 20 **(d)** 17

10. The perpendicular from the centre of a circle to a chord of length 28 cm bisects it. Each half is (in cm): 1

(a) 15 **(b)** 14 **(c)** 13 **(d)** 16

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. AB is a diameter of a circle and C is a point on it with $\angle CAB = 46^\circ$. Find $\angle ABC$. 2

12. An arc of a circle subtends 132° at the centre. Find the angle it subtends at a point on the remaining part of the circle. 2

13. $ABCD$ is a cyclic quadrilateral with $\angle B = 118°$. Find $\angle D$. 2

14. AB is a diameter of a circle and C is a point on it with $\angle CAB = 53°$. Find $\angle ABC$. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A chord of a circle of radius 13 cm is at a distance of 5 cm from the centre. Find the length of the chord. 3

16. Points A and B lie on a circle with centre O and $\angle AOB = 116°$. A point C lies on the major arc. Find $\angle ACB$ and say why $\angle ADB = \angle ACB$ for another point D on the major arc. 3

17. $ABCD$ is a cyclic quadrilateral with $\angle A = 57°$ and $\angle B = 106°$. Find $\angle C$ and $\angle D$. 3

OR

A chord of a circle of radius 13 cm is at a distance of 12 cm from the centre. Find the length of the chord. 3

18. Points A and B lie on a circle with centre O and $\angle AOB = 128°$. A point C lies on the major arc. Find $\angle ACB$ and say why $\angle ADB = \angle ACB$ for another point D on the major arc. 3

OR

$ABCD$ is a cyclic quadrilateral with $\angle A = 95°$ and $\angle B = 68°$. Find $\angle C$ and $\angle D$. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A chord of length 32 cm is at a distance of 12 cm from the centre of a circle. Find the radius. 5

OR

AB is a diameter of a circle and C, D lie on the circle on opposite sides of AB. If $\angle CAB = 15°$ and $\angle DAB = 50°$, find $\angle ABC$ and $\angle ABD$. 5

20. A chord of length 24 cm is at a distance of 16 cm from the centre of a circle. Find the radius. 5

OR

AB is a diameter of a circle and C, D lie on the circle on opposite sides of AB. If $\angle CAB = 38°$ and $\angle DAB = 22°$, find $\angle ABC$ and $\angle ABD$. 5

END OF QUESTION PAPER — **Total: 40 Marks**

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